Ejercicio

Resolver:

$xy''' - y'' = x^2$ con

$y(1) = 0 \; ; \; y'(1) = 1 \; \text{y} \; \; y''(1) = 2$

Notar que es una ecuación diferencial de orden 3, lineal, no homogénea y con coeficientes variables

$xy''' - y'' = 0$

$x\,v' - v = 0$

$x\,v' = v$

$v = y''$

$v' = y'''$

$v' = \frac{v}{x}$ $\quad \frac{dv}{dx} = \frac{v}{x}$ $\quad dv \cdot x = v \cdot dx$

$\frac{dv}{v} = \frac{dx}{x}$

$\frac{1}{v}\,dv = \frac{1}{x}\,dx$

$\int \frac{1}{v}\,dv = \int \frac{1}{x}\,dx$

$\ln|v| = \ln|x| + c$

$e^{\ln|v|} = e^{\ln|x| + c}$

$v = cx \longrightarrow y'' = cx$

$y' = \frac{cx^2}{2} + c_2$ $\quad y = \frac{cx^3}{2 \cdot 3} + c_2 x + c_3$

$y = \frac{c}{6}x^3 + c_2 x + c_3$

$y_h = c_1 x^3 + c_2 x + c_3$

$xy''' - y'' = x^2$ $\quad v = y''$

$x\,v' - v = x^2$ $\quad v' = y'''$

$v' - \frac{1}{x}v = x$

FACTOR INTEGRANTE

$[f \cdot v]' = fv' + f'v$

$\mu \cdot v - \frac{\mu}{x}v = \mu x$ $\quad (\mu \cdot v)' = \mu v' + \mu' v$

$\mu' v = -\frac{\mu}{x}v$ $\quad \mu' = \frac{d\mu}{dx}$

$\mu' = -\frac{\mu}{x}$ $\quad \frac{d\mu}{dx} = -\frac{\mu}{x}$ $\quad \frac{d\mu}{\mu} = -\frac{dx}{x}$

$\frac{1}{\mu}\,d\mu = -\frac{1}{x}\,dx$ $\quad \int \frac{1}{\mu}\,d\mu = \int -\frac{1}{x}\,dx$

$\ln|\mu| = -\ln|x| + c$ $\quad e^{\ln|\mu|} = e^{-\ln|x| + c}$

$|\mu| = e^{-\ln|x|} \cdot e^c$ $\quad |\mu| = \frac{1}{e^{\ln|x|}} \cdot e^c = \frac{e^c}{|x|}$

$|\mu| = \frac{k}{|x|}$ $\quad \mu = \frac{1}{x}$

$v' - \frac{1}{x}v = x$ $\quad \boxed{\frac{1}{x}v' - \frac{1}{x^2}v = 1}$

$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ $\quad f = v$ $\quad g = x$

$f' = v'$ $\quad g' = 1$

$\left(\frac{v}{x}\right)' = 1$ $\quad \left(\frac{v}{x}\right)' = \frac{v'x - v \cdot 1}{x^2} = \frac{v'x}{x^2} - \frac{v}{x^2}$

$\frac{v}{x} = x + c$ $\quad = \frac{v'}{x} - \frac{v}{x^2} = \boxed{\frac{1}{x}v' - \frac{1}{x^2}v}$

$v = x^2 + cx$

$y'' = x^2 + cx$ $\quad y'' = x^2$

$y' = \frac{x^3}{3} = \frac{1}{3}x^3$ $\quad y = \frac{1}{3} \cdot \frac{x^4}{4}$

$y_p = \frac{1}{12}x^4$

$y = c_1 x^3 + c_2 x + c_3 + \frac{1}{12}x^4$

$y(1) = 0 \; ; \; y'(1) = 1 \; \text{y} \; \; y''(1) = 2$

$y' = 3c_1 x^2 + c_2 + \frac{1}{3}x^3$

$y'' = 6c_1 x + x^2$

$c_1 + c_2 + c_3 + \frac{1}{12} = 0$ $\quad 3c_1 + c_2 + \frac{1}{3} = 1$

$6c_1 + 1 = 2$

$6c_1 = 1$

$\begin{cases} c_1 + c_2 + c_3 = -\frac{1}{12} \\ 3c_1 + c_2 = \frac{2}{3} \\ 6c_1 = 1 \end{cases}$

$c_1 = \frac{1}{6}$

$3 \cdot \frac{1}{6} + c_2 = \frac{2}{3}$ $\quad c_2 = \frac{1}{6}$

$\frac{1}{6} + \frac{1}{6} + c_3 = -\frac{1}{12}$ $\quad c_3 = -\frac{5}{12}$

$y = c_1 x^3 + c_2 x + c_3 + \frac{1}{12}x^4$

$y = \frac{1}{6}x^3 + \frac{1}{6}x - \frac{5}{12} + \frac{1}{12}x^4$

$y = \frac{1}{12}x^4 + \frac{1}{6}x^3 + \frac{1}{6}x - \frac{5}{12}$